Time-Decay Attribution Model Using Half-Life in Exponential Decay for E-Commerce Conversion Analysis

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***Abstract*—With the rapid growth of e-commerce platform users, it has become essential for companies to analyze the customer journey leading to a purchase. Users may interact with or be exposed to multiple devices and touchpoints/channels throughout this process. So now, to identify which of those touch-points resulted in a positive conversion, the companies use various types of attribution models. By identifying this, the company then allocates its marketing spend accordingly in the future on the chosen channels to further optimize its sales and customized user experience. Over time, various types of attribution models have been developed, each model having its pros and cons. This study proposes a novel Time-Decay Attribution Model using Half-life in Exponential Decay (TD-HLED). We also provide a comparative analysis with six different Models: First-touch, Markov Chain Model, Linear-Touch, Last-touch, and our proposed model, TD-HLED. Furthermore, extensive experiments have proved that the model attributions in TD-HLED have more accurately reflected how the users have interacted with the marketing than the existing simplistic rule-based architectures.**

***Index Terms*—Conversion, Touch-points/channels, MTA (Multi-touch Attribution), TD-HLED**

I. Introduction

Multi-touch attribution models are a method used to assign weight or credit to each touch-point or channel a customer interacts with on their journey toward the final conversion [1]. For example, let's say that a customer is looking for a new smartphone. The user goes through various tech blogs and smartphone review videos on YouTube, and while watching the video, he's shown a targeted ad from Apple. At first, the user chooses to skip the ad and ignore it. Then, while surfing through Instagram, he sees a native ad in his feed from an iPhone promoted by his favorite Pop-Star Band, which immediately catches his attention, and then he clicks on the link and is redirected to the iPhone's Official Website. He sees the price, chooses wisely not to make an $800 purchase hastily, and closes the site. Finally, he receives an email from iPhone about an ongoing sale and an exclusive discount code, which causes him to make the purchase. This is known as a conversion, and his conversion journey can be shown as:

Organic Search →Instagram →Email

Each of these ads represent a touch-point in the buyer's journey. And each of these touch-points had a certain psychological impact on the customer, but how to determine which one of those touch-points/channels had the bigger influence/weightage over the others becomes the next question. And here comes the attribution models for the rescue. They all have their own style/method of treating and assigning/distributing the weights/credits to all the touch-points. These attribution models give the marketers critical insights into understanding the customer's preferences and psychology. It allows companies to create highly tailored and personalized ads, making them even harder for customers to overlook or decline.

The rest of the paper is organized as follows. Section II and III present the literature review and contribution of this study, respectively. The proposed method is discussed in Section IV. Experimental setup and performance analysis of the proposed model are presented in Section V. Finally, Section VI concludes this study.

II. Related Works

Most of the existing models were built based upon the probabilistic model, last-touch model, attention model, survival theory, dual-attention model, deep neural network with attention model, and further cross-variations among the mentioned models in this field of attribution modeling. These models were built with the core ideology of having a data-driven model that identifies the pattern of the consumer's journey, conversion-delay in the touch-points and accordingly assigns credits to the touch-points, rather than implementing the simple intuition-based rules models like the popular last-touch model where the last touch-point in the consumer's journey is given the highest credit for the conversion [2]. In [3], the authors have proposed a probabilistic multi-touch attribution model which monitors all touch-points/ads the consumer has viewed/clicked and the time at which they viewed/clicked it too, by using the survival theory, Weibull distribution, and hazard rate of conversion. In our proposed model, every channel and click is credited with a value out of 100 on the basis of the half-life equation (i.e., Eq. (1)).

In [4], the authors have proposed a Casual-Attention model to predict channel attribution using users' pre-conversion actions with a deep recurrent neural network architecture. It also minimizes the selection bias in channel assignment across time stamps and touch-points. Rather than using pre-conversion credits, we offer credits to every touch-point, but the credit value reduces as the touch-point gets further and further away from the conversion point.

In [5], the authors have proposed a Deep Neural Network based Attention multi-touch attribution model (DNAMTA) to reduce estimation biases of media effects using users' context information (e.g., user demographics and behavior). In comparison to the complex DNAMTA model, our model assigns credits to every touch-point, but the value reduces with time, which also helps us identify the channels that are least used by users during a period of time.

In [6], an Enhanced Last Touch Interaction (ELTI) model is used to distribute revenue in different marketing channels using synergistic effects and game theory. In [7], the authors have compared two machine learning-based attribution models (i.e., LSTM and SVM) to find a better conversion model.

III. Contributions

Our proposed Time Decay using Half-Life in Exponential Decay(TD-HLED) model can track the least-used and most-used channels in a consumer's converting journey. The model shows an exponential graph of users navigating through different channels, and upon analyzing the graphs of all users, the channels that were visited least frequently and most frequently could be identified. For example, a user navigates through the following channels:
Twitter →Facebook →YouTube →LinkedIn →Amazon →LinkedIn →Amazon(Converted)

The model credits Twitter & Facebook the least as they were visited initially and never visited again before the conversion. So Twitter & Facebook were the least-visited channels, and Amazon & LinkedIn were the most-visited channels. Analyzing the data of mass users shows the least-visited channels (which are not in trend), and running the analysis for a given period of time shows the most valuable channel generating the highest revenue, thus helping the marketing team to allocate the budget for each channel in the future. This assists the company in identifying trending platforms and maximizing profits over a short period, such as quarterly or semi-annually.

IV. Proposed Approach

Our proposed TD-HLED Model allocates more credit to touch points that occur closer in time to the conversion. It assumes that touch-points that are closer in time to the conversion are more likely to have influenced the customer's decision to convert. The closer a touch-point is to the conversion, the more credit it is given. The further away a touch-point is from the conversion, the less credit it is given. Thus, this model can be used to identify the touch-points that are most effective at driving conversions in a specific time frame.

To distribute the credits amongst the channels/touch-points, we used the half-life equation 1. Typically, half-life denotes the time a quantity/substance requires to reduce to half of its size/value. In our context, this half-life is used to decrease/distribute the credits assigned to a touch-point/channel after a set amount of time(i.e., seven days, here). Equation (1) computes the half-life for conversion credit distribution.

$$y = 2^{\frac{-x}{7}} \tag{1}$$

where, y represents the credit assigned by the TD-HLED Model to each touch-point/channel.x is the no. of days the referral happened prior to the conversion. In Eq. (1), x is divided by 7 as the half-life is considered 7. By that, what we mean is suppose Channel-1 was visited 7 days before Channel-2, then Channel-1 will receive half the credit assigned to Channel-2 for the conversion.

Equation (1) has been derived from equation (2) by taking $N_0 = 1$, $N(t) = y$, $t_{\frac{1}{2}} = 7$ and $t = -x$. So, Eq. (2) represents more generalized formula for Half-life [8].

$$N(t) = N_0 2^{\frac{t}{t_{\frac{1}{2}}}} \tag{2}$$

where, $N(t)$ denotes the quantity that has not decayed after a time t. N_0 denotes the initial amount before measuring decay. t denotes the time span an interaction is away from conversion. $t_{\frac{1}{2}}$ denotes half-life which is the time it takes for a quantity to reduce to half its original value.

V. Experimental Setup and Result

We have conducted a series of experiments to evaluate the proposed TD-HLED model. First, we introduce the experimental setup, including the data set and different attribution models (First Touch, Last Touch, Linear Touch, and Markov Model). Second, we interpret the parameters of the proposed new model (TD-HLED model) to get a better conversion rate and reduce delay in conversion. Third, we measure the performance of the proposed model of conversion prediction, reduce delay conversion, and compare it with the existing models. Lastly, we discuss the characteristics of the graph in the proposed model.

A. Dataset

For this analysis, we used the dataset[1] present in Kaggle. It shows data from digital marketing encountered in a production environment. The data set contains 586,000 marketing touch-points from July (2018), comprising 240,000 unique customers who generated approximately 18,000 conversions.

A more detailed description of the features of the dataset is as follows.

- *Cookie*: Anonymous customer ID enabling us to track the progression of a given customer.
- *Timestamp*: Date and time when the visit took place.

[1]https://www.kaggle.com/code/hughhuyton/multitouch-attribution-modelling

- *Interaction*: Categorical variable indicating the type of interaction that took place.
- *Conversion*: Boolean variable indicating whether a conversion took place or not (0 or 1).
- *Conversion Value*: Value of the potential conversion event (revenue).
- *Channel* : The marketing channel that brought the customer to our site like (Instagram, Facebook, Web Page, Twitter etc.).

Using all the given data, we could re-construct the full path taken by a customer while buying a product online; the chronological sequence of clicking advertisements, channels used, and conversion rate can be determined. The initial data set is raw data saved in a CSV file containing a lot of noise. To make a fair comparison between different models, we have cleaned the raw data by following the method and then applied our proposed Algorithm 1:

1) *Removing irreverent data like punctuation, website address, etc.,*
2) *Tokenization of the dataset into table.*
3) *Converting the time stamps from string variable to date and time format, which can be interpreted by Python code.*
4) *Sorting the data into groups according to customer ID.*

Algorithm 1: Evaluation of gap between touch-points/channels for credit-distribution

Input : Gap (in days) between the conversion touch-point and various touch-point/channels the users went through([30, 22, 14, 8, 0])

Output: The credit-attribution for conversion of all the touch-points the user went through.

Dataset comprising of gap(in days) of different users and their touch-points

for *each i (channel: facebook, instagram, ·) in the data* **do**

$y_1 = 100 * 2^{\frac{-x_1}{7}}$;

$y_2 = 100 * 2^{\frac{-x_2}{7}}$;

$y_3 = 100 * 2^{\frac{-x_3}{7}}$;

/* y_i = credits given, x_i = day gap */

end

Assigning a new matrix (z) with credit values;

Taking transpose of the credit dataset to create comparison graph;

Constructing TD-HLED graph;

Figure 1 shows the time Decay using half-life in exponential decay. It also shows the credit for different users per channel conditions.

B. Result and Analysis

1) Single-Touch Attribution Model: It's the easiest model available out there to implement.

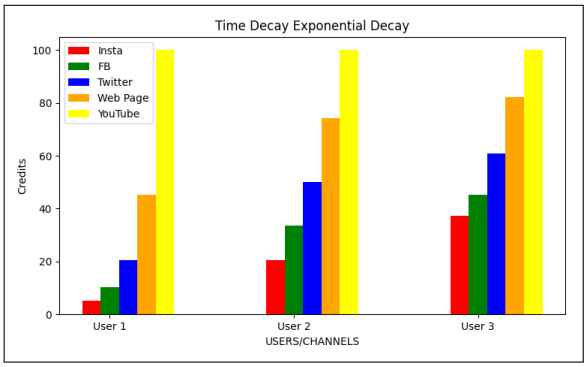


Fig. 1. Time Decay using Half-Life in Exponential Decay

a) First-Touch Attribution Model (FTAM): It attributes credit for a conversion (e.g., a sale or a sign-up) to the first touch-point (e.g., the first ad or website visited) that led to the conversion [9]. This model assumes that the first touch-point is the most important driving factor in the conversion process and that all other touch-points have no or minimal impact (Fig. 2).

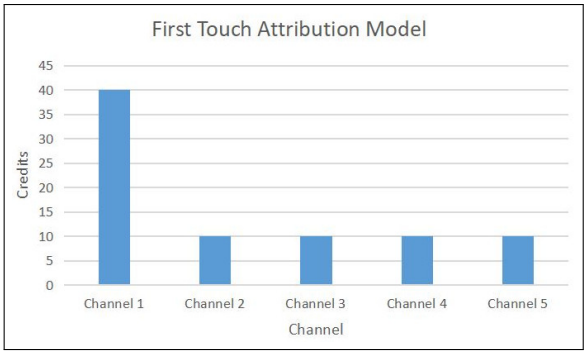


Fig. 2. First-Touch Attribution Model (FTAM).

b) Last-Touch Attribution Model (LTAM): It attributes credit for a conversion to the last touch-point before the conversion occurs [10], [11]. This model assumes that the last touch-point is the most important driving factor in the conversion process and that all other touch-points have no or minimal impact (Fig. 3).

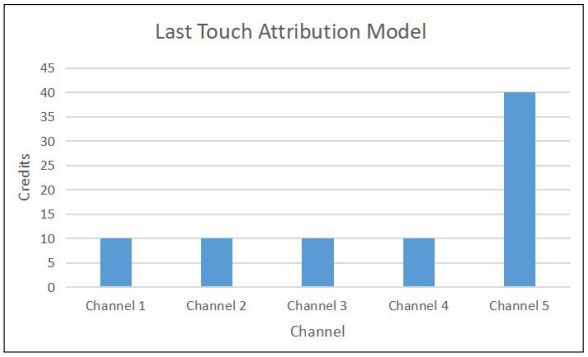


Fig. 3. Last-Touch Attribution Model (LTAM).

2) Multi-Touch Attribution Model:

a) Linear Attribution Model: It attributes credit for a conversion equally among all touch-points in the conversion path [12]. This model assumes all touch-points have contributed equally to the conversion, and the goal is to give credit to all the touch-points that contributed to the sale (Fig. 4.)

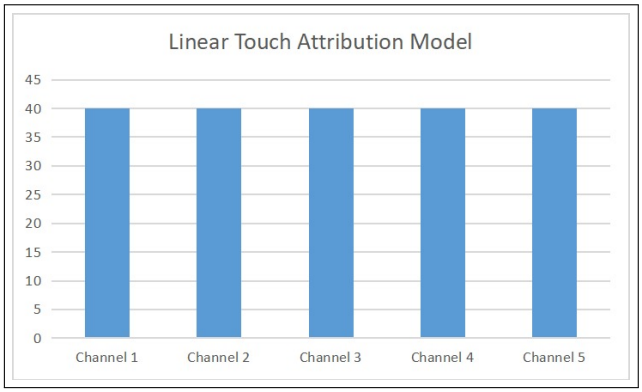


Fig. 4. Linear Touch Attribution Model

b) U-Shaped Model: It attributes more credit to the first and last touch-points in the conversion path, while distributing less credit to the touch-points in between [13]. This model assumes that the first touch-point is responsible for initially attracting the customer and the last touch-point is responsible for closing the sale, and all the touch-points in between are less important in the conversion process (Fig. 5.)

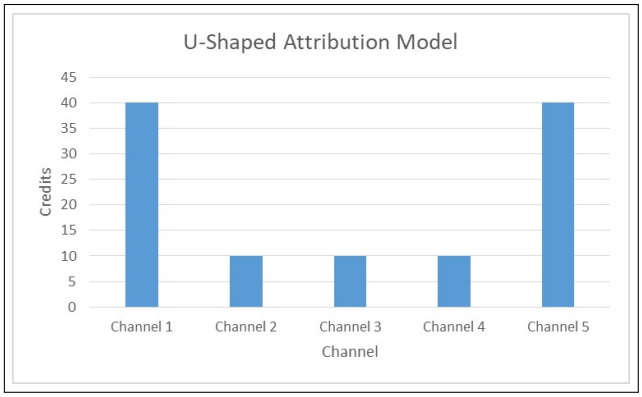


Fig. 5. U-Shaped Attribution Model

c) W-Shaped Model: It attributes most of the credits to the first, the last, and the middle touch-points in the conversion path [14]. This model assumes that different touch-points play different roles in the conversion process, and all of them are important. The first touch-point is responsible for attracting the customer, the middle touch-points are responsible for nurturing and keeping the customer engaged, and the last touch-point is responsible for closing the sale (Fig. 6.)

d) Markov's Chain Attribution Model: It attributes credits to touch-points based on the analysis of "how the removal of that touch-point will affect the likelihood of conversion. This model assumes that the probability of transition to a

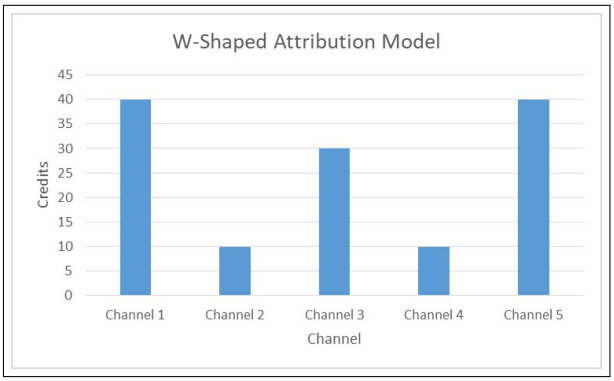


Fig. 6. W-Shaped Model

touch-point solely depends on the current touch-point. Sometimes, this model gives the same attribution as linear attribution model (Fig. 7.)

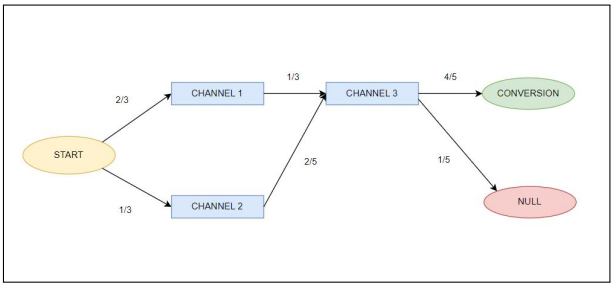


Fig. 7. Markov's Chain Attribution Model

C. Comparative Analysis of TD-HLED with Other MTA Models

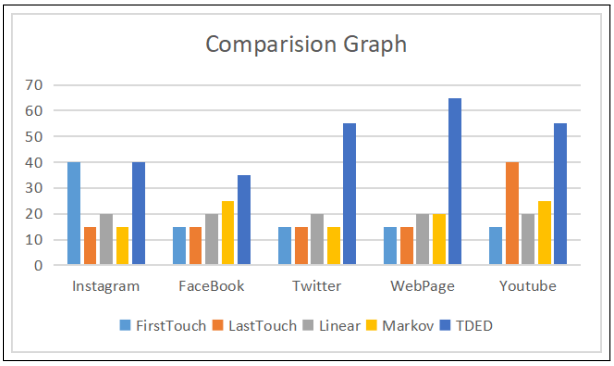


Fig. 8. Comparative analysis between different models.

The bar chart (Fig. 8.) shows how many conversions were attributed to each channel for each model (e.g., FirstTouch, LastTouch, etc.). Analyzing the graph, specifically the TD-HLED Model in comparison to the other methods, you can gain insights as to the relative importance of different marketing channels. For the first touch, last touch and linear touch models, Facebook and Paid Search are the most important channels driving conversions while Instagram and Online

Display are the least important. However, according to the TD-HLED Model, web page channel is far more important to our conversions than what the simple attribution models suggest - indeed according to the probabilistic model it is in fact the most important channel. A comparison analysis of TD-HLED model with FTAM, LTAM, Linear model, W-shape, and U-shape are shown in Fig. 9, Fig. 10, Fig. 11, Fig. 12, and Fig. 13, respectively.

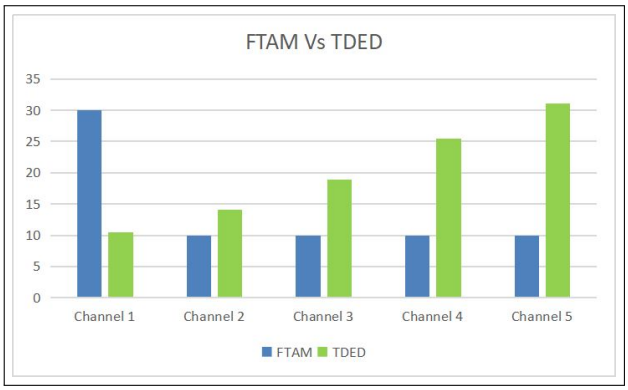


Fig. 9. FTAM vs TD-HLED

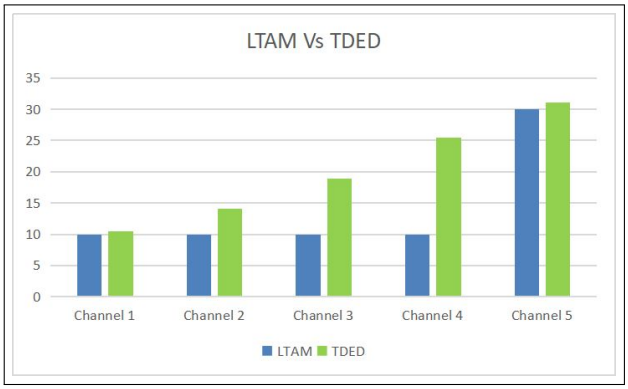


Fig. 10. LTAM vs TD-HLED

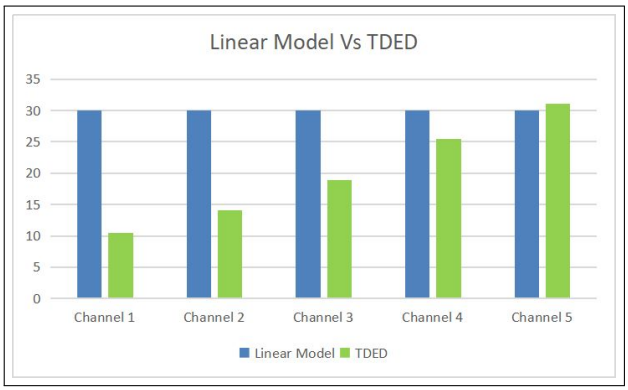


Fig. 11. Linear Model vs TD-HLED

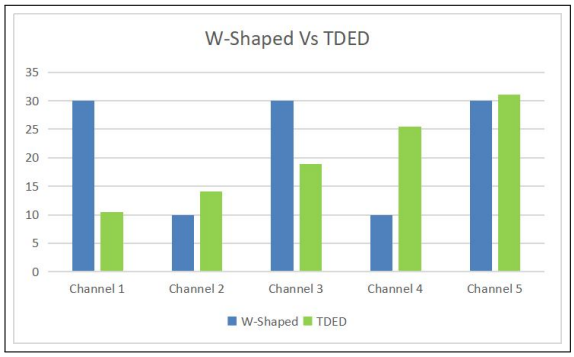


Fig. 12. W-Shaped vs TD-HLED

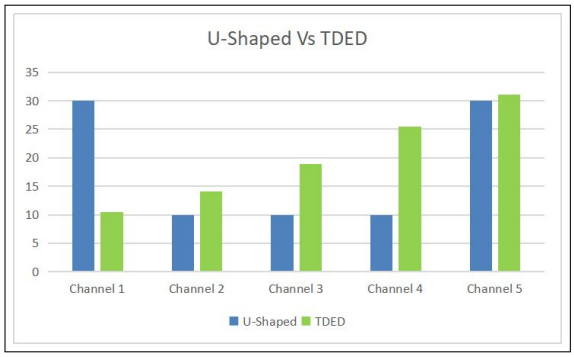


Fig. 13. U-Shaped vs TD-HLED

D. Case Study

Let's assume a customer sees an Ad for Nike Air Jordans on YouTube, but despite their fleeting interest, they decide to suppress the urge due to the $170 price tag. After three days, while scrolling through a midnight flash sale on Amazon, he comes across the same product at a 17% discount; he clicks on it but, seeing the $140 and being in a half-asleep mood, decides to wishlist that item. But now, this product is more firmly lodged in his mind. After three more days, while scrolling through his daily Facebook feed, he again stumbles upon yet another enticing ad by an influencer who also shares a limited edition discount coupon. He notes down the code and goes back to his work. After two days, he finally decides to make the purchase, so he goes directly to the website and completes the transaction.

The Conversion Path is: Youtube (8 days ago) →Amazon (5 days ago) →Facebook (2 days ago) →Direct visit (0 days ago).

Table I shows the computed credits for each touch-point.

Figure 14 shows the credit conversion for each Channel. The credit scores for different channels are as follows.

- Youtube: 15.71%
- Amazon: 21.14%
- Facebook: 28.46%
- Direct Visit: 34.69%

Thus, we can observe that the maximum credit was attributed to a Direct Visit (being the closest to conversion). Now let's assume the customer bought the shoes for $135,

TABLE I
CREDITS FOR EACH TOUCH-POINT

Channel	x	y	$\bar{y} = y/\sum y$	Credit (%) = $100 \times \frac{y}{\sum y}$
Youtube	8	0.452861832	0.157096195	15.7096195
Amazon	5	0.609506827	0.211435799	21.14357991
Facebook	2	0.820335356	0.284571483	28.45714828
Direct visit	0	1	0.346896523	34.68965231
–	–	$\sum y = 2.882704015$	$\sum \bar{y} = 1$	–

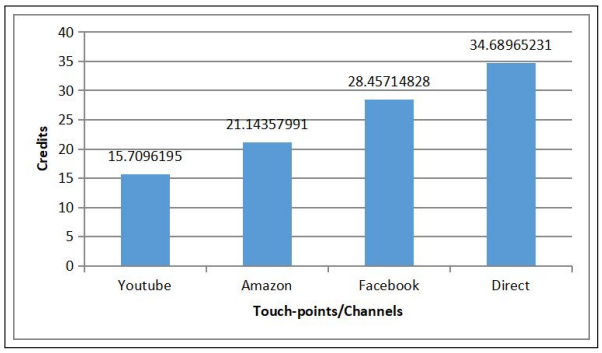


Fig. 14. TD-HLED Model for case study.

and it took \$60 for Nike to manufacture those shoes; then the profit earned by this sale is \$75. Then, according to the credit distribution, Table II indicates the influence/impact of each channel for this sale.

TABLE II
INFLUENCE/IMPACT OF EACH CHANNEL FOR THIS SALE

Channels	Revenue Generated per Channel = Profit x Credit Attributed
YouTube	\$75 x 15.71% = \$11.78
Amazon	\$75 x 21.14% = \$15.86
Facebook	\$75 x 28.46% = \$21.35
Direct Visit	\$75 x 34.69% = \$26.02

Now, assuming that the cost per click/view of
YouTube Ads: \$0.03
Amazon Ads: \$6
Facebook Ads: \$0.94

Then, by observing the revenue generated per channel/touchpoint, we observe that the lower money spent on Facebook Ads actually drove more money/sales than the expensive Amazon Ads. Thus, this would help the marketing team to accordingly allocate/modify the budget in the future.

VI. CONCLUSION

The primary contribution of this paper is our proposed Time-Decay Attribution Model using Half-life in Exponential Decay (TD-HLED) for conversion credit attribution, which considers both the conversion rate of the consumer and the conversion delay. It can provide the marketers much finer and potential insights into the contributions provided by different advertising channels to the eventual conversions of consumers.

The TD-HLED model is fitted to the observed data (conversion rate and conversion delay) and can be applied to conversion prediction, which is of great significance for advertisers to plan and operate their marketing tactics. The TD-HLED model has been evaluated on a large real-world dataset. The extensive experiments have proved its superior effectiveness in both conversion prediction and attribution analysis when compared to the existing models.

In this paper, we set the half-life to 7 days, but in the future, we would like to build a customized half-life for the exponential decay function whose value can change based on the dataset used. This way, the model will provide a more realistic representation of the customer's journey, reflecting how all touchpoints and channels contributed to the conversion.

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